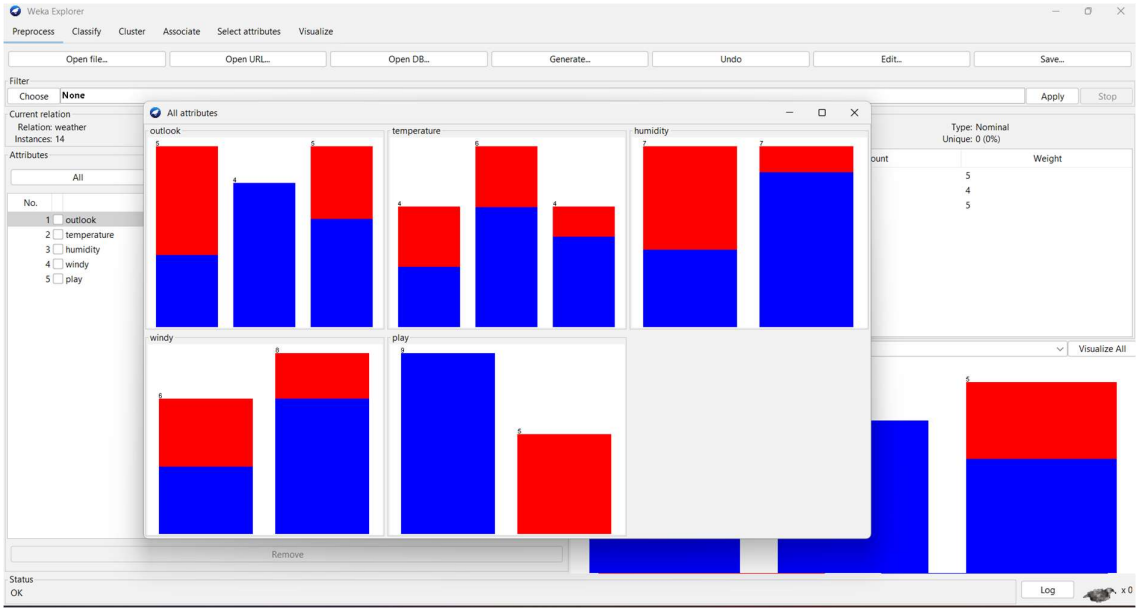
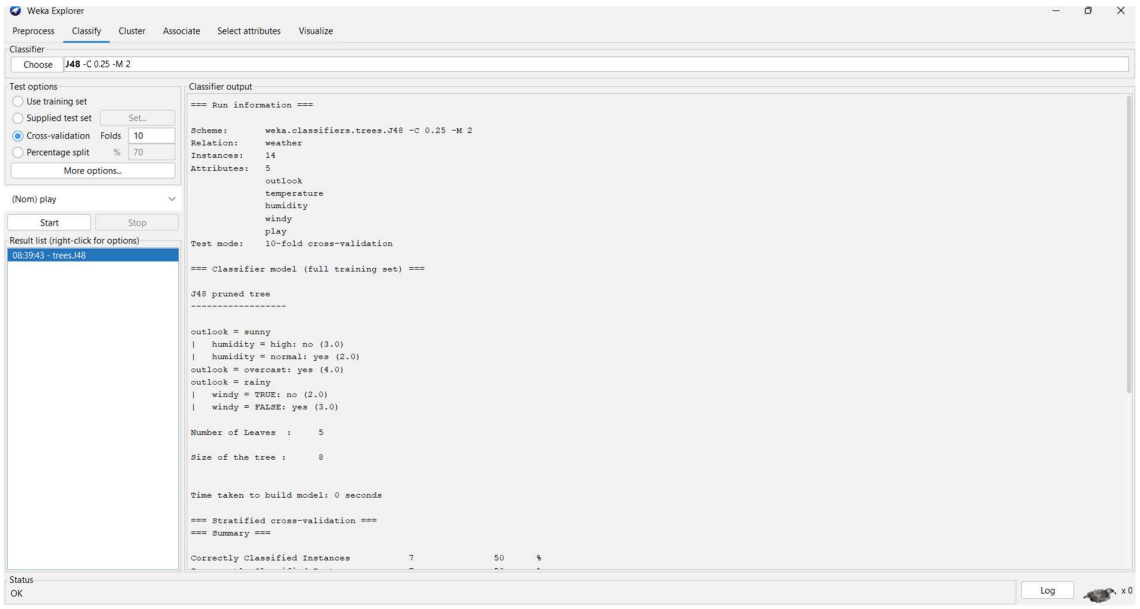
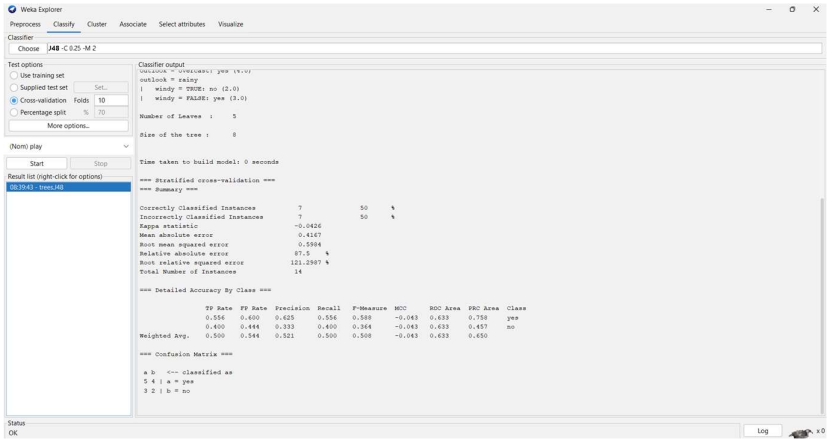
K. Navadeep Reddy

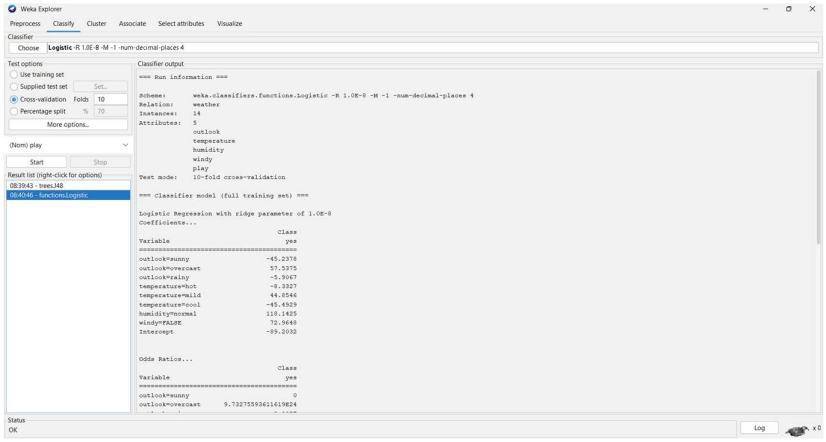
192311305

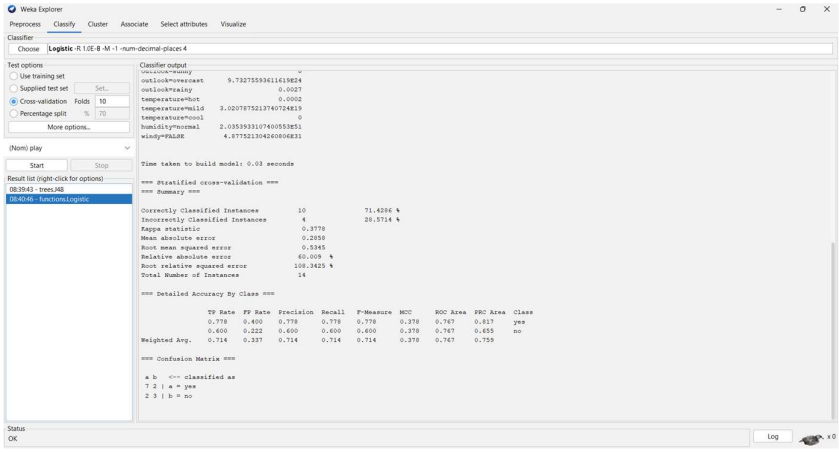
Experiment 26






Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Classifier
Choose J48 -C 0.25 -M 2
Test options
Use training set
Supplied test set Set...
Cross-validation Folds 10
Percentage split % 70
More options...
(Nom) play
Start Stop
Result list (right-click for options)
08:39:43 - trees.J48
Classifier output
outlook = rainy
| windy = TRUE: no (2.0)
| windy = FALSE: yes (3.0)
Number of Leaves : 5
Size of the tree : 8
Time taken to build model: 0 seconds
=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 7 50 %
Incorrectly Classified Instances 7 50 %
Kappa statistic -0.0426
Mean absolute error 0.4167
Root mean squared error 0.5904
Relative absolute error 87.5 %
Root relative squared error 121.2987 %
Total Number of Instances 14
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
0.556 0.600 0.625 0.556 0.588 -0.043 0.633 0.758 yes
0.400 0.444 0.333 0.400 0.364 -0.043 0.633 0.457 no
Weighted Avg. 0.500 0.544 0.521 0.500 0.508 -0.043 0.633 0.650
=== Confusion Matrix ===
a b <-- classified as
5 4 | a = yes
3 2 | b = no
Status
OK
Log
x 0


Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Classifier
Choose Logistic -R 1.0E-8 -M -1 -num-decimal-places 4
Test options
Use training set
Supplied test set Set...
Cross-validation Folds 10
Percentage split % 70
More options...
(Nom) play
Start Stop
Result list (right-click for options)
08:39:43 - trees.J48
08:40:46 - functions.Logistic
Classifier output
=== Run information ===
Scheme: weka.classifiers.functions.Logistic -R 1.0E-8 -M -1 -num-decimal-places 4
Relation: weather
Instances: 14
Attributes: 5
outlook
temperature
humidity
windy
play
Test mode: 10-fold cross-validation
=== Classifier model (full training set) ===
Logistic Regression with ridge parameter of 1.0E-8
Coefficients...
Class
Variable yes
outlook=sunny -45.2378
outlook=overcast 57.5375
outlook=rainy -5.9067
temperature=hot -8.3327
temperature=mild 44.8546
temperature=cool -45.4929
humidity=normal 118.1425
windy=FALSE 72.9648
Intercept -89.2032
Odds Ratios...
Class
Variable yes
outlook=sunny 0
outlook=overcast 9.732755936116198E24
Status
OK
Log
x 0


Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Classifier
Choose Logistic -R 1.0E-8 -M -1 -num-decimal-places 4
Test options
Use training set
Supplied test set Set...
Cross-validation Folds 10
Percentage split % 70
More options...
(Nom) play
Start Stop
Result list (right-click for options)
08:39:43 - trees.J48
08:40:46 - functions.Logistic
Classifier output
outlook=overcast 9.732755936116198E24
outlook=rainy 0.0027
temperature=hot 0.0002
temperature=mild 3.020787521374072481 9
temperature=cool 0
humidity=normal 2.035393310740055385 51
windy=FALSE 4.877521304260806E31
Time taken to build model: 0.03 seconds
=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 10 71.4286 %
Incorrectly Classified Instances 4 28.5714 %
Kappa statistic 0.3778
Mean absolute error 0.2858
Root mean squared error 0.5345
Relative absolute error 60.009 %
Root relative squared error 108.3625 %
Total Number of Instances 14
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
0.778 0.400 0.778 0.778 0.778 0.378 0.767 0.817 yes
0.600 0.222 0.600 0.600 0.600 0.378 0.767 0.655 no
Weighted Avg. 0.714 0.337 0.714 0.714 0.714 0.378 0.767 0.759
=== Confusion Matrix ===
a b <-- classified as
7 2 | a = yes
2 3 | b = no
Status
OK
Log
x 0